ORIGINAL ARTICLE

doi: 10.2176/jns-nmc.2025-0370

Surgical Management Strategy for Large Cystic Vestibular Schwannoma Based on the Cyst Classification

Masahide MATSUDA,[1] Akinari YAMANO,[1] and Eiichi ISHIKAWA[1]

[1]*Department of Neurosurgery, Institute of Medicine, University of Tsukuba, Tsukuba, Ibaraki, Japan*

Abstract

Cystic vestibular schwannomas often grow more rapidly and adhere more strongly to the facial nerves and brainstem than do solid tumors. Patients with large cystic tumors may experience sudden clinical deterioration during the preoperative waiting period; furthermore, it is important to carefully consider whether to dissect the cyst wall from adjacent structures. Accordingly, we aimed to clarify the influence of different cyst types on surgical strategies. We included 19 patients with large cystic vestibular schwannomas (extrameatal diameter >30 mm) who underwent microsurgical resection. Tumors were classified using the Piccirillo et al. system based on the cyst location and wall thickness. We compared the incidence of sudden clinical deterioration and surgical outcomes according to the cyst types. Peripherally located thin-walled cysts (type B) were significantly more likely to cause sudden clinical deterioration than were centrally located thick-walled cysts (type A). In addition, when the thin cyst wall was firmly adhered to the facial nerve or brainstem, a conservative surgical strategy was applied, with the wall being intentionally left in place rather than attempting forceful dissection. This approach achieved excellent facial nerve preservation but relatively decreased the extent of resection. Taken together, these findings suggest that large cystic vestibular schwannomas with peripherally located thin-walled cysts have a high risk of rapid clinical deterioration and may require early surgical intervention. It is important to adapt the dissection strategy according to cyst wall thickness to achieve optimal postoperative preservation of the facial nerve.

Keywords: vestibular schwannoma, cyst classification, clinical deterioration, extent of resection, facial nerve function

Introduction

Vestibular schwannoma (VS) is the most common benign tumor arising in the cerebellopontine angle. It originates from Schwann cells that form the myelin sheath of the vestibular nerve. Generally, VSs can be classified into solid and cystic types, with most cases being solid. Cystic tumors account for approximately 5.7%-10% of all VSs.[1,2] Cystic VSs (CVSs) tend to exhibit more rapid growth and stronger adhesion to the facial nerve and brainstem than do solid tumors.[3,4] Accordingly, cystic tumors are associated with a lower rate of postoperative preservation of the facial nerve than that of solid tumors.

The Piccirillo et al.[5] classification system is used to categorize cystic tumors on the basis of their intratumoral location and cyst wall thickness. On the basis of this system, type A tumors are centrally located with thick walls, whereas type B tumors are eccentrically located with thin walls (Figure 1).

Patients with large cystic VSs can present with sudden clinical deterioration during the preoperative waiting period; accordingly, it is important to carefully consider the timing of the surgical intervention. Moreover, given that the cyst wall often firmly adheres to the facial nerve and brainstem, aggressive dissection for total resection may increase the risk of postoperative complications, including facial nerve palsy.

Therefore, it is important to clarify 2 key issues in the surgical management of large cystic VSs: whether early surgical intervention is warranted and the extent of dissection of the cyst wall from the facial nerve and brainstem. Accordingly, we aimed to clarify the influence of different

Received November 5, 2025; Accepted December 19, 2025

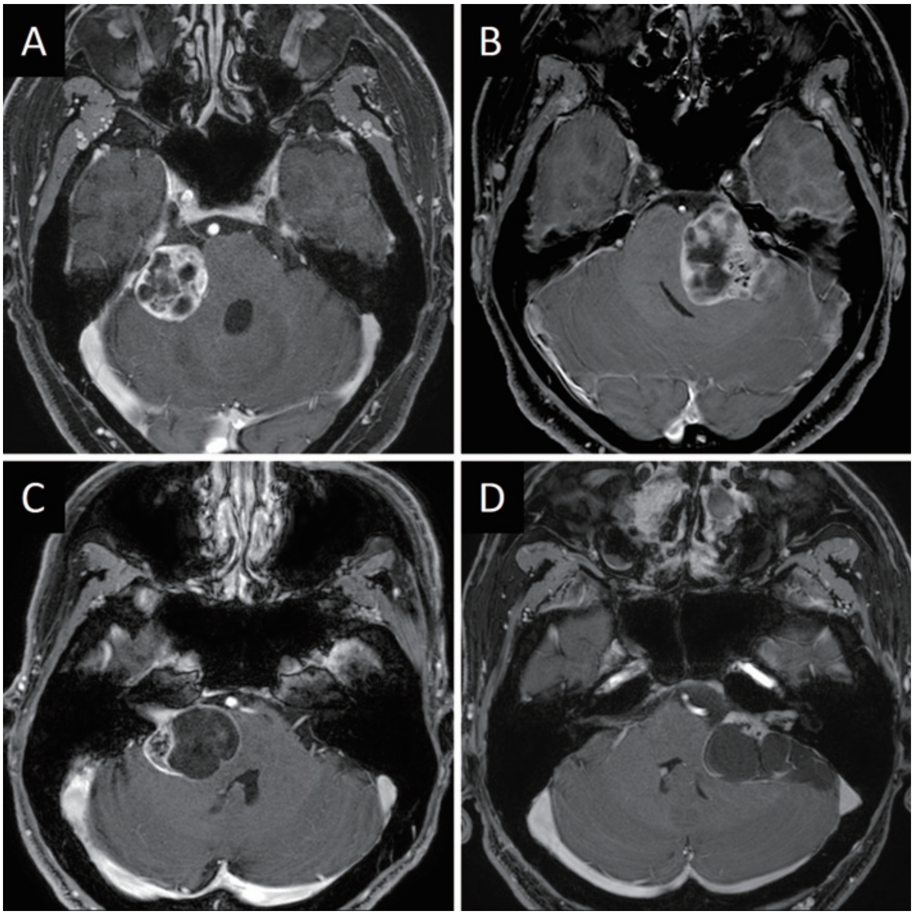

Figure 1 Representative MRI findings of the Piccirillo classification system.
(A, B) Type A cystic vestibular schwannomas with centrally located, thick-walled cysts.
(C, D) Type B cystic vestibular schwannomas with peripherally located, thin-walled cysts.
MRI: magnetic resonance imaging

cyst types on surgical strategies with respect to these key issues.

Materials and Methods

We retrospectively included patients with large CVSs who underwent surgery performed by the same neurosurgeon (M.M.) at the University of Tsukuba Hospital between January 2017 and September 2024. Among 78 consecutive patients, 20 were classified as having large CVSs. Among them, we excluded 1 patient with neurofibromatosis type 2; accordingly, 19 patients were included in the final analysis. A cystic tumor was defined as one in which cystic components comprised ≥50% of the total tumor volume; furthermore, it was considered large when the extrameatal diameter exceeded 30 mm.

Cystic morphology was classified on the basis of the Piccirillo et al.[5] system. As aforementioned, type A tumors are centrally located with thick walls and may present as multiple small cysts (A1), multiple moderately sized cysts (A2), or a single large cyst (A3). Contrastingly, type B tumors are peripherally located with thin walls, with the cysts extending predominantly in the anterior (B1), medial (B2), or posterior (B3) directions, or showing a combined pattern (B4). In addition, magnetic resonance imaging (MRI) findings were evaluated to assess the maximum tumor diameter, number of cysts (monocystic or polycystic), and presence of a fluid-fluid level suggestive of a previous intratumoral hemorrhage.[6]

Sudden clinical deterioration during the period from the date of the first visit to our institution to the date of surgery was defined as a marked worsening of cranial nerve symptoms, cerebellar dysfunction, or headache. The extent of tumor resection was categorized as near total resection (NTR), subtotal resection (STR), or partial resection (PR). NTR was defined as a minimal residual tumor with no clearly enhancing lesions on postoperative contrast-enhanced MRI. STR was defined as a residual tumor ≤5% of the preoperative volume. PR was defined as a residual tumor >5% of the preoperative volume. Facial nerve function was evaluated using the House-Brackmann (HB) grading system at hospital discharge and 1 postoperative year,[7]

Table 1 Clinical and Radiological Characteristics of 19 Patients with Large Cystic Vestibular Schwannoma

Characteristic	Value
Age (years)	
mean±SD	65.0±8.9
range	49-79
Sex	
male	11 (57.9%)
female	8 (42.1%)
Tumor maximum diameter (mm)	
mean±SD	38.2±6.6
range	30-53
Cyst classification	
type A	10 (52.6%)
type B	9 (47.4%)
Cyst multiplicity	
monocystic	7 (36.8%)
polycystic	12 (63.2%)
Fluid-fluid level	5 (26.3%)
Preoperative waiting period (days)	
median (IQR)	27 (17-34)
Sudden clinical deterioration	6 (31.6%)

IQR: interquartile range; SD: standard deviation

with HB grades I-II and III-VI indicating good and poor facial nerve function, respectively.

All procedures were performed using the retrosigmoid approach, with opening of the internal auditory canal to allow resection of the intrameatal portion. Moreover, continuous facial nerve monitoring and transcranial facial nerve-evoked potential monitoring were performed. If the cyst wall was thick, subcapsular dissection was maximized to preserve the facial nerve and brainstem by only leaving the tumor capsule comprising the vestibular nerve fibers and perineurium.[8] If the cyst wall was thin and strongly adhered to the surrounding structures, it was intentionally left in place to avoid facial nerve or brainstem injury.

This study was approved by the institutional ethics committee (approval number: R01-202), and an opt-out method for informed consent was granted, considering the retrospective nature of this analysis.

All statistical analyses were performed using SPSS version 29.0. Continuous variables, including age and tumor diameter, were expressed as mean ± standard deviation and compared using the t-test; contrastingly, the preoperative waiting period was expressed as median (interquartile range) and analyzed using the Mann-Whitney U test. Categorical variables were analyzed using the chi-squared test. Statistical significance was set at $p < 0.05$.

Results

Among the 19 included cases, 10 and 9 were classified as type A and B, respectively. Table 1 summarizes the clinical and radiological characteristics. The patients ranged in age from 49 to 79 years, with a mean age of 65.0 years. There were 11 men and 8 women. The maximum extrameatal tumor diameter ranged from 30 to 53 mm (mean, 38.2 mm). In addition, 7 and 12 tumors were monocystic and polycystic, respectively. Five cases showed fluid-fluid level suggestive of intratumoral hemorrhage. The median preoperative waiting period, defined as the period from the date of the first visit to our institution to the date of surgery, was 27 days (17-34). Six patients (31.6%) experienced sudden clinical deterioration during this period.

Figure 2 shows a representative case of sudden clinical deterioration. A 66-year-old man with a type B cystic tumor initially presented with right-sided hearing loss, facial sensory disturbance, and facial weakness (HB grade III); furthermore, he could independently walk despite mild imbalance. After 20 days, at the time of admission for surgery, his facial palsy worsened to HB grade IV. In addition, he could not walk without assistance and had repeated falls, which caused dental fractures. Gadolinium-enhanced MRI revealed enlargement of the cystic component with increased brainstem compression.

Comparative analysis revealed that the cyst type was significantly associated with sudden clinical deterioration (Table 2). Specifically, sudden clinical deterioration was significantly more frequent in patients with type B tumors than in those with type A tumors. Contrastingly, sudden clinical deterioration was not significantly associated with other factors, including maximum tumor diameter, cyst multiplicity, presence of a fluid-fluid level suggestive of a previous intratumoral hemorrhage, age, or sex. The preoperative waiting period was shorter in patients who experienced sudden clinical deterioration, partly because some underwent earlier surgery owing to symptom worsening than did those without deterioration (13 days [5.5-28.8] vs. 28 days [22-41]). Notably, the preoperative waiting period did not significantly differ according to cyst type (type A, 27 days [20.8-46.8] vs. type B, 27 days [10.5-34.0]).

Table 3 summarizes the surgical outcomes according to cyst type. In the type A group, NTR and STR were achieved in 5 patients each. In the type B group, NTR and STR were achieved in 1 and 8 cases, respectively. This indicates that for patients with type B tumors, the extent of resection was intentionally reduced to leave the cyst wall adherent to the facial nerve or brainstem.

Preoperative facial nerve palsy was observed in 3 patients in the type A group (all HB grade II) and 5 patients in the type B group (2 and 3 with HB grades II and III-VI, respectively). There was postoperative worsening of facial nerve function in 2 and 3 patients in the type A and B groups, respectively. At 1 postoperative year, mild residual

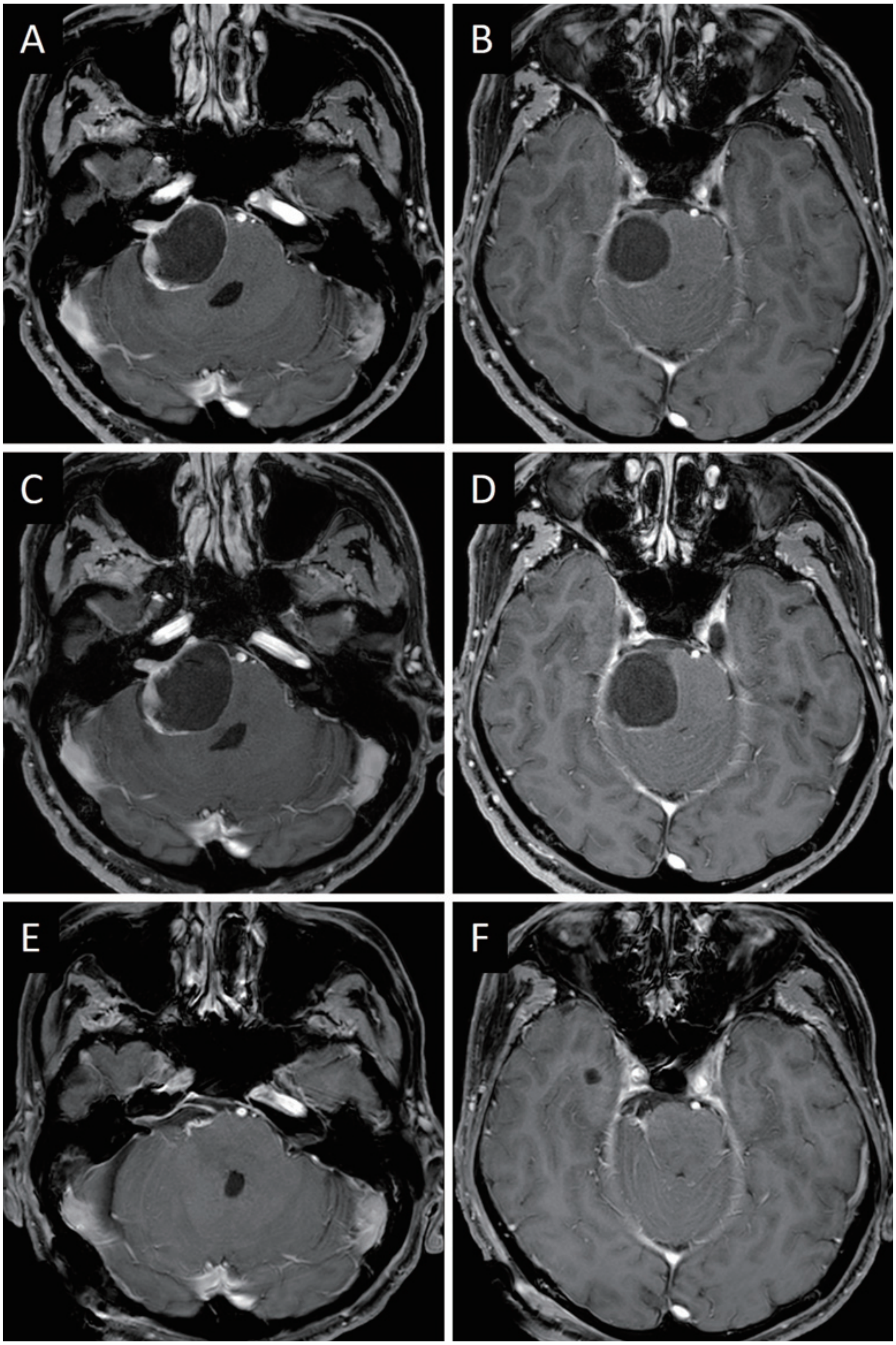

Figure 2 MRI findings in a representative case of type B cystic vestibular schwannoma with sudden clinical deterioration. Gadolinium-enhanced T1-weighted axial MRI.
(A, B) Preoperative MRI for surgical examination. The cystic tumor compressed the brainstem.
(C, D) Images obtained on admission for surgery after sudden clinical deterioration. No intratumoral hemorrhage was observed; however, the cyst was enlarged, causing increased brainstem compression.
(E, F) MRI obtained at 1 postoperative week. Subtotal resection was achieved, leaving the cyst wall adherent to the brainstem and facial nerves.
MRI: magnetic resonance imaging

Table 2 Analysis of Sudden Clinical Deterioration in Patients with Large Cystic Vestibular Schwannoma

	Sudden clinical deterioration		p Value
	Yes (n = 6)	No (n = 13)	
Age (years)	64.5±12.1	65.2±7.6	0.87
Sex			
Male	4 (66.7%)	7 (53.8%)	0.60
Female	2 (33.3%)	6 (46.2%)	
Tumor maximum diameter (mm)	40.0±8.7	37.3±5.7	0.43
Cyst classification			
Type A	1 (16.7%)	9 (69.2%)	0.03
Type B	5 (83.3%)	4 (30.8%)	
Cyst multiplicity			
Monocystic	3 (50.0%)	4 (30.8%)	0.42
Polycystic	3 (50.0%)	9 (69.2%)	
Fluid-fluid level			
Yes	2 (33.3%)	3 (23.1%)	0.64
No	4 (66.7%)	10 (76.9%)	
Preoperative waiting period (days)	13 (5.5-28.8)	28 (22-41)	0.04

Age is presented as mean ± standard deviation.
Preoperative waiting period is presented as median (interquartile range).

Table 3 Surgical Outcomes According to Cyst Classification

	Type A (n = 10)	Type B (n = 9)	p Value
Extent of resection			
NTR	5 (50%)	1 (11.1%)	0.07
STR	5 (50%)	8 (88.9%)	
Preoperative facial nerve function			
HB grade I	7 (70%)	4 (44.4%)	
HB grade II	3 (30%)	2 (22.2%)	0.14
HB grade III-VI	0	3 (33.3%)	
Change in facial nerve function at discharge compared with preoperative status			
Worsened	2 (20%)	3 (33.3%)	0.51
Unchanged/improved	8 (80%)	6 (66.7%)	
Facial nerve function at 1 year postoperatively			
HB grade I	9 (90%)	9 (100%)	
HB grade II	1 (10%)	0 (%)	0.33
HB grade III-VI	0 (%)	0 (%)	

HB: House–Brackmann; NTR: nearly total resection; STR: subtotal resection

facial weakness (HB grade II) persisted in 1 patient in the type A group, whereas in all patients in the type B group, complete recovery was achieved. Consequently, all included patients, regardless of cyst type, had good facial nerve function (HB grades I-II) at the 1-year follow-up.

Discussion

Our findings indicated that in large CVSs with an extrameatal diameter >30 mm, tumors characterized by peripherally located and thin-walled cysts had a significantly high risk of sudden clinical deterioration during the preoperative waiting period. Moreover, our findings suggest that adapting the surgical dissection strategy according to the cyst wall thickness allows satisfactory preservation of postoperative facial nerve function.

CVSs exhibit distinct clinical behaviors from those of solid tumors, including a faster growth rate, larger tumor size at diagnosis, and more rapid progression of clinical symptoms.[4,9,10] Sudden clinical deterioration in patients with CVSs can be attributed to sudden cyst expansion or intratumoral hemorrhage.[10,11] Predicting symptom progression is crucial when planning surgery for large cystic VSs because cases at risk for extremely rapid worsening may require semi-urgent surgical intervention. However, to our knowledge, the predictive factors for sudden clinical deterioration remain unclear. Our findings indicated that cyst type based on the Piccirillo et al.[5] classification was a useful predictor. Contrastingly, tumor size and the presence of a fluid-fluid level suggestive of a previous intratumoral hemorrhage were not correlated with clinical deterioration. These findings indicate that although new intratumoral bleeding may trigger a sudden clinical decrease, a history of hemorrhage is not predictive of future deterioration. Among patients with type B cysts, 4 of the 5 cases without evidence of hemorrhage showed rapid cyst enlargement. This suggests that type B cysts may have an intrinsic tendency toward accelerated cyst expansion compared with type A cysts. Although the underlying mechanism remains unclear, cyst enlargement in VSs may be attributed to progressive fluid accumulation driven by osmotic imbalance and/or leakage of serum proteins due to blood-tumor barrier disruption.[9] The thin cyst walls in type B tumors may contribute to these processes by facilitating osmotic fluid accumulation.

CVSs have a lower rate of postoperative preservation of the facial nerve function than do solid tumors.[3,4,12] Moreover, postoperative outcomes have been shown to differ according to the Piccirillo et al.[5] classification. Nair et al.[13] analyzed 64 patients with large CVSs (>2.7 cm; 43 and 21 with type A and B, respectively), with complete tumor resection being achieved in 83% of these patients. However, at 6 postoperative months, severe facial nerve palsy (HB grades IV-VI) was observed in 80% of all cases (74% and 90% of type A and B cases, respectively). On the basis of these results, the authors suggested that in type B cysts, especially anterior or medial cysts adhering to the facial nerve, subtotal resection may be preferable to preserve facial nerve function. Huo et al.[14] reviewed 188 cystic VSs of various sizes (79 type A, 109 type B) and similarly reported that dissecting a thin cyst wall from the facial nerve was technically difficult owing to strong adhesions. In type B tumors, the extent of resection was deliberately limited to preserving the facial nerve, which initially generated lower preservation rates at discharge (HB grades I-II in 27%) than those in type A tumors (42%). However, this conservative approach proved beneficial over time, with facial nerve function improving during follow-up. Moreover, there was no significant between-group difference in the preservation rates at 1 postoperative year (34% in type B vs. 47% in type A).

Our surgical policy for large cystic VSs is to achieve maximal tumor resection sufficient to avoid retreatment while preventing permanent facial nerve palsy. In the dissection of solid tumors, even large tumors, we routinely use subcapsular dissection, leaving the capsule comprising vestibular nerve fibers and perineurium on the facial nerve side to preserve its integrity. Although subcapsular dissection may occasionally be challenging in large tumors, it facilitates both maximal resection and functional preservation when feasible. In our series, thick-walled type A tumors often allowed partial or extensive subcapsular dissection, similarly to solid tumors. Conversely, in thin-walled type B tumors, it was generally impossible to enter the subcapsular plane between the cyst wall and facial nerve. In such cases, we intentionally left the cyst wall adhering to the facial nerve or brainstem without attempting a forceful dissection, which yielded excellent postoperative preservation of the facial nerve. Consistent with the recommendations of Piccirillo et al.[5] and Nair et al.,[13] we consider that in cases with thin-walled cysts adherent to the facial nerve or brainstem, limiting resection to a planned subtotal level is essential to preserve long-term facial nerve function.

This study had some limitations. First, selection bias may exist given the retrospective study design. Second, the small sample size may limit the generalizability of our findings. Third, the relatively short follow-up period did not allow the assessment of long-term recurrence, especially in patients who underwent subtotal resection. Nevertheless, to our knowledge, this is the first study to identify the factors associated with sudden clinical deterioration in patients with large CVSs. Our findings provide valuable insights into determining surgical timing and strategies for such challenging tumors. Large-scale prospective multicenter studies are warranted to validate these findings and establish evidence-based management guidelines.

Conclusions

Our findings suggest that large CVSs with peripherally located thin-walled cysts have a high risk of sudden clinical deterioration and may require early surgical intervention. It is important to adapt the dissection strategy according to cyst wall thickness to achieve optimal postoperative preservation of the facial nerve.

Conflicts of Interest Disclosure

All authors have no conflict of interest.

Funding

This work was supported in part by Grants-in-Aid for Scientific Research (KAKENHI) from the Japan Society for the Promotion of Science awarded to Masahide Matsuda (Number JP 24K12281).

References

1) Fundová P, Charabi S, Tos M, et al. Cystic vestibular schwannoma: surgical outcome. J Laryngol Otol. 2000;114(12):935-9. doi: 10.1258/0022215001904653
2) Jones SE, Baguley DM, Moffat DA. Are facial nerve outcomes worse following surgery for cystic vestibular schwannoma? Skull Base. 2007;17(5):281-4. doi: 10.1055/s-2007-986436
3) Thakur JD, Khan IS, Shorter CD, et al. Do cystic vestibular schwannomas have worse surgical outcomes? Systematic analysis of the literature. Neurosurg Focus. 2012;33(3):E12. doi: 10.3171/2012.6.FOCUS12200
4) Wu X, Song G, Wang X, et al. Comparison of surgical outcomes in cystic and solid vestibular schwannomas: a systematic review and meta-analysis. Neurosurg Rev. 2021;44(4):1889-902. doi: 10.1007/s10143-020-01400-5.
5) Piccirillo E, Wiet MR, Flanagan S, et al. Cystic vestibular schwannoma: classification, management, and facial nerve outcomes. Otol Neurotol. 2009;30(6):826-34. doi: 10.1097/MAO.0b013e3181b04e18
6) Xia L, Zhang H, Yu C, et al. Fluid-fluid level in cystic vestibular schwannoma: a predictor of peritumoral adhesion. J Neurosurg. 2014;120(1):197-206. doi: 10.3171/2013.6.JNS121630
7) House JW, Brackmann DE. Facial nerve grading system. Otolaryngol Head Neck Surg. 1985;93(2):146-7. doi: 10.1177/019459988509300202
8) Sasaki T, Shono T, Hashiguchi K, et al. Histological considerations of the cleavage plane for preservation of facial and cochlear nerve functions in vestibular schwannoma surgery. J Neurosurg. 2009;110(4):648-55. doi: 10.3171/2008.4.17514
9) Kameyama S, Tanaka R, Kawaguchi T, et al. Cystic acoustic neurinomas: studies of 14 cases. Acta Neurochir (Wien). 1996;138(6):695-9. doi: 10.1007/BF01411474
10) Sinha S, Sharma BS. Cystic acoustic neuromas: surgical outcome in a series of 58 patients. J Clin Neurosci. 2008;15(5):511-5. doi: 10.1016/j.jocn.2007.01.007
11) Charabi S, Tos M, Thomsen J, et al. Cystic vestibular schwannoma--clinical and experimental studies. Acta Otolaryngol Suppl. 2000;543:11-3. doi: 10.1080/000164800453810-1
12) Stastna D, Mannion R, Axon P, et al. Facial nerve function outcome and risk factors in resection of large cystic vestibular schwannomas. J Neurol Surg B Skull Base. 2022;83(suppl 2):e216-24. doi: 10.1055/s-0041-1725028
13) Nair S, Baldawa SS, Gopalakrishnan CV, et al. Surgical outcome in cystic vestibular schwannomas. Asian J Neurosurg. 2016;11(3):219-25. doi: 10.4103/1793-5482.145359
14) Huo Z, Zhang Z, Huang Q, et al. Clinical comparison of two subtypes of cystic vestibular schwannoma: surgical considerations and outcomes. Eur Arch Otorhinolaryngol. 2016;273(12):4215-23. doi: 10.1007/s00405-016-4149-4

Corresponding author: Masahide Matsuda, MD, PhD
Department of Neurosurgery, Institute of Medicine, University of Tsukuba, 1-1-1 Tennodai, Tsukuba, Ibaraki 305-8575, Japan.
e-mail: m-matsuda@md.tsukuba.ac.jp